UVFaceFusion: Fast Multi-view Topologically Consistent Face Reconstruction in the Wild via UV-space Neural Fusion

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Fig. 1. Given casual in-the-wild multi-view captures as input, with four of the sixteen images shown here, our method reconstructs a high-fidelity mesh with consistent topology in under three seconds. It generalizes robustly across capture environments, lighting conditions, facial expressions, and ethnic groups, enabling everyday users to bring themselves into the digital world through casual scanning.

Reconstructing high-fidelity facial geometry with an assigned topology is essential for digital avatar creation and animation, yet existing automated methods often trade off geometric fidelity and in-the-wild generalization. We present UVFaceFusion, a feed-forward framework for multi-view, fixed-topology face reconstruction from daily images. Our key idea is to replace heuristic topological optimization with learnable neural fusion in a canonical UV space. Given multi-view images, we first obtain dense point maps and facial UV correspondences of each view using VGGT and Pixel3DMM, respectively. Then, the view-specific point maps are lifted into the canonical UV domain and fused with a novel mask-aware neural fusion network. The network predicts a complete UV-space point map, from which a fixed-topology mesh is directly sampled. Although trained only on Ava-256, UVFaceFusion generalizes well to multiple public benchmarks and in-the-wild captures, benefiting from its canonical UV-space geometry-to-geometry fusion that reduces dependence on dataset-specific appearance and capture conditions. Experiments on various benchmarks show that UVFaceFusion achieves state-of-the-art reconstruction accuracy while reconstructing a mesh from 16 input views in less than 3 seconds on a single RTX 4090. Code is available at https://github.com/grignarder/UVFaceFusion.

CCS Concepts: • **Computing methodologies** → **Reconstruction**.

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1 INTRODUCTION

Establishing facial geometry with an assigned topology is a fundamental requirement for digital avatar creation [Alexander et al. 2013, 2009], as dense surface correspondences are required to facilitate rigging, animation, and texture transfer. Conventional paradigms generally separate the process into face geometry reconstruction followed by template fitting [Egger et al. 2020; Riviere et al. 2020]. However, ensuring high-fidelity output usually necessitates extensive manual labor for the second step, such as landmark supervision and meticulous parameter tuning, which restricts the scalability of these pipelines for widespread consumer applications. On the other hand, 3D Morphable Model (3DMM)-based methods [Deng et al. 2019; Feng et al. 2021] can be used for automatic template fitting but are inherently limited by the low-dimensional expression space, failing to capture person-specific geometric traits. To pursue higher fidelity, multi-view learning approaches [Li et al. 2024, 2021] employ neural networks to infer geometry directly from images. However, these methods essentially couple geometry reconstruction with topology building, making them heavily reliant on studio-captured datasets for end-to-end training. Consequently, they struggle to generalize to in-the-wild scenarios where lighting and camera configurations deviate from the controlled training environment.

To resolve this generalization-fidelity trade-off, VGGTFace [Ming et al. 2026] proposes a *decoupled* paradigm that decomposes topologically consistent reconstruction into two stages: general geometric reconstruction and topological fusion. Given smartphone-captured multi-view face images in the wild, VGGTFace leverages a robust, general-purpose estimator (VGGT [Wang et al. 2025]) for geometry prediction (in terms of view-specific point maps), which achieves high generalization across diverse environments. Then, it applies a heuristic Bundle Adjustment (BA) [Triggs et al. 1999] process to integrate these view-specific predictions into a unified, topologically consistent mesh. However, this hand-crafted BA optimization lacks semantic awareness to distinguish noise in geometry predictions from actual facial features, often resulting in an over-smooth surface or topological artifacts that fall short of the precision required by professional pipelines.

To close this gap, we present UVFaceFusion, which evolves the decoupled paradigm by replacing the rigid, heuristic BA with a learnable neural fusion module. To inherit the high generalization ability of general-purpose estimators, we operate exclusively on geometric point maps, effectively shielding the fusion process from the detrimental effects of lighting variations and camera distortion that are pre-processed by the underlying VGGT backbone. Specifically, we utilize the predicted UV correspondence by Pixel3DMM [Giebenhain et al. 2026] to warp view-specific 3D coordinates into a canonical UV space, thereby transforming the irregular multi-view fusion problem into a structured, pixel-aligned feature aggregation task. Then, we employ a shared-weight encoder to extract deep geometric feature maps from these aligned point maps, followed by a mask-aware feature fusion layer that dynamically aggregates cross-view information to resolve occlusions and prediction uncertainties. Finally, a decoder reconstructs the refined, topologically consistent facial geometry in the UV domain. By focusing on geometry-to-geometry mapping, our framework demonstrates exceptional data efficiency; it can be supervised on limited studio datasets, such as Ava-256 [Martinez et al. 2024], yet generalizes seamlessly to unconstrained in-the-wild faces. Notably, given 16 input views, UVFaceFusion reconstructs a fixed-topology face mesh in less than 3 seconds on a single RTX 4090, achieving a 3× inference speedup over VGGTFace while setting new benchmarks in both qualitative fidelity and quantitative accuracy.

In conclusion, our main contributions include:

- We present an efficient feed-forward system that reconstructs face meshes with fixed topology from 16 in-the-wild images in less than 3 seconds on a single RTX 4090, with a 3× speedup over VGGTFace and state-of-the-art accuracy across multiple benchmarks.
- We introduce a canonical UV-space fusion representation that lifts VGGT point maps into a shared facial parameterization using Pixel3DMM correspondences, converting fixed-topology face reconstruction into a structured UV point-map prediction problem.
- We propose a mask-aware neural fusion network that first extracts per-view geometric features with a shared encoder and then aggregates them in feature space, enabling robust fusion of noisy and partial multi-view observations.

2 RELATED WORK

2.1 Monocular Face Reconstruction

Reconstructing 3D facial geometry from a single image is inherently ill-posed, and a large body of work addresses this ambiguity with 3D Morphable Model (3DMM) priors [Blanz and Vetter 1999; Egger et al. 2020; Li et al. 2017; Zollhöfer et al. 2018]. Given an input image, these methods either solve an analysis-by-synthesis fitting problem [Dib et al. 2021; Giebenhain et al. 2026; Thies et al. 2016] or regress the parameters of a 3DMM/FLAME model with a neural network [Daněček et al. 2022; Deng et al. 2019; Feng et al. 2021; Retsinas et al. 2024; Wang et al. 2024b; Zielonka et al. 2022]. Recent methods further improve robustness to challenging expressions and in-the-wild images by using denser supervision such as facial-part segmentation [Wang et al. 2024b], neural rendering objectives [Retsinas et al. 2024], or foundation-model features for screen-space normal and UV prediction [Giebenhain et al. 2026]. While these approaches produce meshes with a consistent template topology, their final geometry is still largely governed by the underlying parametric face space and the ambiguity of a single observation.

To improve geometric expressiveness, many works augment the parametric model with per-vertex offsets, UV displacement maps, or learned nonlinear corrective spaces [Dib et al. 2024; Feng et al. 2021; Han et al. 2023; Lei et al. 2023; Tewari et al. 2019; Yang et al. 2020]. These extensions recover more local details, yet they typically treat high-frequency geometry as residuals on top of a coarse 3DMM fit. Another line of work reduces the dependence on 3DMM parameters by directly predicting dense geometry or correspondences in image/UV space [Feng et al. 2018; Guo et al. 2023; Richardson et al. 2017; Sela et al. 2017; Yun et al. 2025; Zhang et al. 2021]. For example, Sela et al. [2017] estimate a UV coordinate image and a depth map and then fit a template mesh to obtain topologically consistent geometry, while later methods predict UV position maps, implicit fields, or dense UV-depth maps. These representations are closely related to ours because they expose dense correspondences and geometric maps rather than only low-dimensional coefficients. However, monocular methods must hallucinate self-occluded regions from learned priors, and per-image predictions do not exploit multi-view observations that are available in our setting.

In contrast, our method treats monocular face reconstruction modules as providers of partial per-view cues instead of final reconstruction engines. We use Pixel3DMM to obtain pixel-aligned UV maps and combine them with VGGT point maps [Wang et al. 2025]; after warping them into a shared UV domain, our networks learn to fuse masked partial observations from multiple views and predict a complete UV-space point map. This design preserves the fixed topology required by downstream facial animation pipelines, while freeing the final geometry from the low-dimensional 3DMM space and reducing the hallucination burden inherent to single-image reconstruction.

2.2 Multi-view Face Reconstruction

Multi-view observations provide complementary geometric cues and largely reduce the ambiguity of monocular reconstruction. Traditional production pipelines usually first reconstruct an unstructured facial scan using multi-view stereo or photogrammetry, and

then register a template mesh to obtain dense correspondence [Fyffe et al. 2017; Riviere et al. 2020]. Although such pipelines can achieve high fidelity, they typically require controlled capture setups, accurate calibration, and substantial manual post-processing. Some learning-assisted methods formulate multi-view face reconstruction as 3DMM regression or non-rigid multi-view stereo [Bai et al. 2020; Wu et al. 2019]. By enforcing cross-view consistency, these methods improve over monocular reconstruction, but their geometry is still constrained by parametric or adaptive face spaces and thus has limited ability to represent person-specific details.

Recent works directly infer topologically consistent meshes from calibrated multi-view images. ToFu [Li et al. 2021] predicts template vertices with volumetric sampling, ReFA [Liu et al. 2022] performs recurrent feature alignment in the UV space to recover production-grade face assets, and TEMPEH [Bolkart et al. 2023] learns to predict dense-correspondence head meshes with view- and surface-aware volumetric feature fusion. GRAPE [Li et al. 2024] further improves the generalization of such systems to unseen camera arrays by using visual-hull-based localization and visibility-aware feature aggregation. Very recently, MOCHI [Filntisis et al. 2026] removes the need for precomputed registered meshes during training by using differentiable point-map and normal-map losses with a topology-regularizing inverse-kinematics branch. Despite their impressive accuracy, these methods are mainly designed for calibrated studio captures and often rely on carefully processed scans or registered meshes for supervision. Their generalization to casual in-the-wild captures remains challenging.

Another relevant line of work predicts dense screen-space geometric maps that can be used for multi-view integration. WarpHE4D [Yun et al. 2025] predicts a dense 4D head map, including per-pixel UV coordinates, depth, and confidence, enabling multi-view point-cloud fusion and downstream template registration. This representation is closely related to ours because it exposes dense correspondences between image pixels and a shared UV domain. However, WarpHE4D predicts each view independently and leaves the fusion of incomplete and noisy observations to downstream processing. In contrast, our method learns to aggregate partial UV-space observations from multiple views into a complete fixed-topology facial geometry.

Beyond facial geometry, related multi-view aggregation problems have also been studied in 3D texture generation. MVPaint [Cheng et al. 2025] synchronizes diffusion across multiple rendered views, completes uncovered surface regions through spatial-aware 3D inpainting, and subsequently performs texture super-resolution and seam smoothing in UV space. More recent methods avoid view-space fusion by generating surface attributes in native 3D representations. NaTex [Lai et al. 2026] models texture as a dense color point cloud and performs latent color diffusion, while TEXTRIX [Zeng et al. 2026] represents color and other surface attributes using a sparse volumetric attribute grid. These methods synthesize appearance for a given 3D surface, whereas our method reconstructs facial geometry by fusing predicted geometric observations from real multi-view images. Nevertheless, they highlight a related representation choice between reconciling partial view-dependent signals in a parameterized surface domain and modeling attributes directly in 3D. In our setting, all outputs share the same fixed FLAME topology and semantically aligned UV parameterization; therefore, canonical UV space provides direct cross-view and cross-subject correspondence and converts geometric fusion into a regular 2D completion problem. Native 3D attribute representations are complementary alternatives for general objects without such a shared semantic parameterization.

Recent general-purpose geometry models have established dense point maps as a flexible representation for feed-forward 3D reconstruction. DUSt3R [Wang et al. 2024a] formulates pairwise reconstruction as point-map regression, while MASt3R [Leroy et al. 2024] augments this formulation with dense matching descriptors. VGGT [Wang et al. 2025] further supports joint inference over multiple views and directly predicts camera parameters and pixel-aligned point maps in a common coordinate frame, making it a convenient geometry backbone for our multi-view pipeline.

Within multi-view facial geometry reconstruction, the most closely related work to ours is VGGTFace [Ming et al. 2026], which introduces a decoupled paradigm for in-the-wild multi-view facial geometry reconstruction. It first leverages VGGT [Wang et al. 2025] to predict view-specific point maps and then injects topology with Pixel3DMM [Giebenhain et al. 2026]. The resulting per-view topological point clouds are fused by a topology-aware Bundle Adjustment process. While this design inherits the strong generalization ability of VGGT, its fusion stage is a hand-crafted optimization based on reprojection and Laplacian regularization. Such an optimization lacks semantic awareness to distinguish noisy tracks from genuine facial structures, and may therefore over-smooth high-frequency details or produce local artifacts. Our method inherits the robustness of foundation-model point maps for in-the-wild geometry estimation, while replacing heuristic test-time fusion with a learned reconstruction framework in canonical UV space. Specifically, we lift view-specific point maps into a shared facial UV parameterization and aggregate the resulting masked partial observations through neural fusion. This geometry-to-geometry formulation enables efficient fixed-topology reconstruction and better preserves person-specific facial shape, expressions, and local details.

3 METHOD

As shown in Figure 2, given multi-view face images, we first use VGGT to estimate pixel-aligned point maps and Pixel3DMM to predict dense UV correspondences. We lift each view's point map into the canonical facial UV domain, yielding partial UV-space geometry observations. A mask-aware neural fusion network then aggregates these partial observations in feature space and predicts a complete UV-space point map. Finally, the fixed-topology mesh is sampled from the UV-space point map.

3.1 Preliminaries

Given a set of multi-view face images $\mathcal{I} = \{I_i\}_{i=1}^{N}$, our goal is to reconstruct a 3D face mesh with a fixed topology. Instead of directly regressing mesh vertices, we represent the output geometry as a dense point map in a canonical facial UV space. The final mesh is obtained by sampling this UV-space point map according to the UV parameterization of a template mesh.

Template mesh and UV parameterization. Let

$$\mathcal{M}_0 = (\bar{\mathbf{V}}, \mathbf{F}, \bar{\mathbf{U}}) \tag{1}$$

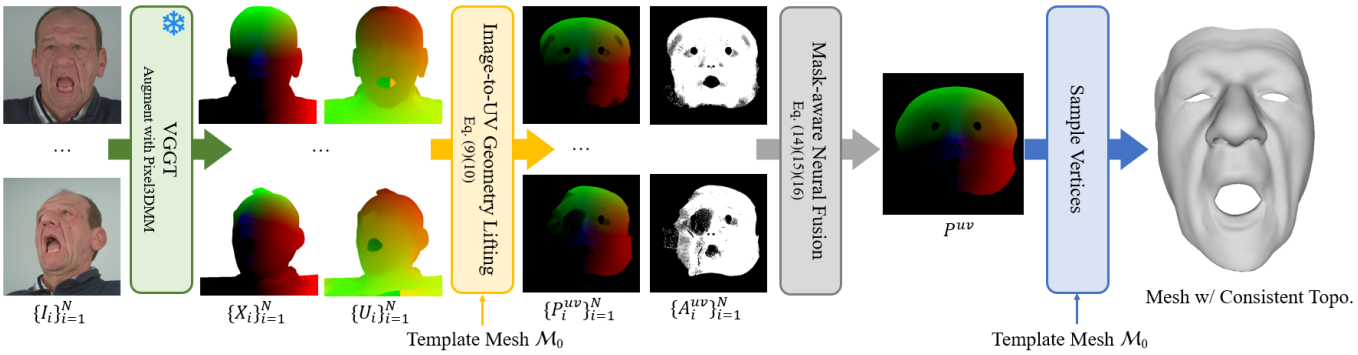

Fig. 2. **Pipeline overview.** Given multi-view face images $\{I_i\}_{i=1}^N$, we use frozen pretrained models to predict pixel-aligned point maps $\{X_i\}_{i=1}^N$ and dense UV correspondences $\{U_i\}_{i=1}^N$. Through image-to-UV geometry lifting and canonicalization with the template mesh $\mathcal{M}_0$, each view is converted into a partial canonical UV-space point map $\{P_i^{uv}\}_{i=1}^N$ with a valid mask $\{A_i^{uv}\}_{i=1}^N$. Our mask-aware neural fusion network aggregates these partial observations and predicts a complete UV-space point map P^{uv}. Finally, we sample mesh vertices from P^{uv} using the template UV coordinates and reuse the template connectivity to obtain a mesh with consistent topology. The snowflake indicates frozen pretrained modules.

denote the canonical template mesh, where $\bar{\mathbf{V}} \in \mathbb{R}^{M\times 3}$ are the template vertices, $\mathbf{F}$ are the fixed triangle faces, and $\bar{\mathbf{U}} = \{\bar{\mathbf{u}}_j\}_{j=1}^M$ are the corresponding vertex UV coordinates with $\bar{\mathbf{u}}_j \in [0,1]^2$. The topology $\mathbf{F}$ is shared by all reconstructed meshes. Our network predicts a UV-space point map

$$\mathbf{P}^{uv} \in \mathbb{R}^{S\times S\times 3}, \tag{2}$$

where each valid UV texel stores a 3D point on the facial surface in the canonical coordinate system. Given $\mathbf{P}^{uv}$, we obtain the fixed-topology mesh by sampling the point map at the template UV coordinates $\bar{\mathbf{U}}$.

VGGT predictions. We use VGGT [Wang et al. 2025] to obtain dense multi-view geometric observations and camera parameters. Given the input images $\{I_i\}_{i=1}^N$, VGGT predicts

$$\{\mathbf{X}_i, \mathbf{K}_i, \mathbf{E}_i\}_{i=1}^N = \Phi_{\text{VGGT}}(\{I_i\}_{i=1}^N), \tag{3}$$

where $\mathbf{X}_i \in \mathbb{R}^{H\times W\times 3}$ is a pixel-aligned point map for the i-th view, $\mathbf{K}_i \in \mathbb{R}^{3\times 3}$ is the camera intrinsic matrix, and $\mathbf{E}_i \in \mathbb{R}^{3\times 4}$ is the camera extrinsic matrix. Following VGGT, the predicted point maps are represented in a common coordinate frame defined by the first camera.

Dense facial correspondence. To establish dense correspondence between image pixels and the canonical facial UV domain, we use a pretrained Pixel3DMM model [Giebenhain et al. 2026]. For each input image I_i, Pixel3DMM predicts a UV map:

$$(\mathbf{U}_i) = \Phi_{\text{P3D}}(I_i), \tag{4}$$

where

$$\mathbf{U}_i \in [0,1]^{H\times W\times 2}. \tag{5}$$

For each image pixel $\mathbf{p} = (x, y)$, $\mathbf{U}_i(\mathbf{p}) = (u, v)$ indicates its corresponding location in the canonical UV domain. The UV maps are used to lift VGGT point maps into the canonical UV space.

3.2 Image-to-UV Geometry Lifting

The point maps predicted by VGGT provide dense 3D observations in image space, while our target representation is defined in the canonical facial UV domain. We therefore lift each per-view point map into UV space using the dense UV correspondence predicted by Pixel3DMM. This converts multi-view image observations into a set of partial UV-space point maps that share the same facial parameterization.

Forward splatting to UV space. Let Ω_i denote the set of valid face pixels in the i-th image, and let $\Omega_U = \{0, \dots, S-1\}^2$ denote the discrete UV texel grid. For each valid image pixel $\mathbf{p} \in \Omega_i$, the UV map $\mathbf{U}_i$ provides its continuous UV coordinate

$$\mathbf{U}_i(\mathbf{p}) = (u_i(\mathbf{p}), v_i(\mathbf{p})) \in [0,1]^2. \tag{6}$$

We forward-splat the corresponding 3D point $\mathbf{X}_i(\mathbf{p})$ to its neighboring UV texels using bilinear weights. For a UV texel $\mathbf{q} = (q_x, q_y) \in \Omega_U$, we define

$$\omega_i(\mathbf{p}, \mathbf{q}) = \left[1 - |q_x - (S-1)u_i(\mathbf{p})|\right]_+ \left[1 - \left|q_y - (S-1)v_i(\mathbf{p})\right|\right]_+, \tag{7}$$

where $[\cdot]_+ = \max(\cdot, 0)$. The accumulated splatting weight is

$$\mathbf{W}_i^{uv}(\mathbf{q}) = \sum_{\mathbf{p}\in\Omega_i} \omega_i(\mathbf{p}, \mathbf{q}). \tag{8}$$

The partial UV-space point map is then computed by weighted averaging:

$$\tilde{\mathbf{P}}_i^{uv}(\mathbf{q}) = \frac{\sum_{\mathbf{p}\in\Omega_i} \omega_i(\mathbf{p}, \mathbf{q})\mathbf{X}_i(\mathbf{p})}{\mathbf{W}_i^{uv}(\mathbf{q}) + \epsilon}. \tag{9}$$

We also obtain a UV-space valid mask by thresholding the accumulated weight:

$$\tilde{\mathbf{A}}_i^{uv}(\mathbf{q}) = \mathbb{1}\left[\mathbf{W}_i^{uv}(\mathbf{q}) > \tau_w\right]. \tag{10}$$

This lifting step produces a partial UV observation $(\tilde{\mathbf{P}}_i^{uv}, \tilde{\mathbf{A}}_i^{uv})$ for each view. Regions that are invisible, self-occluded, or outside the facial area remain invalid in the UV domain. Thus, the multi-view reconstruction problem is converted into completing and fusing a set of partial UV-space geometric observations.

Canonicalization. Although VGGT represents all views in a common coordinate frame, this frame is tied to the input camera system rather than the canonical template, and the resulting point maps may still contain residual camera and geometry errors. For each template vertex j, we sample the partial UV-space point map $\widetilde{\mathbf{P}}_i^{uv}$ within a local neighborhood centered at its template UV coordinate $\bar{\mathbf{u}}_j$ using masked interpolation. We retain the vertex only when this neighborhood contains sufficient valid support according to $\widetilde{\mathbf{A}}_i^{uv}$, yielding a set of supported vertex indices $\mathcal{J}_i$. Since each sampled point $\hat{\mathbf{v}}_{i,j}$ and the corresponding canonical vertex $\bar{\mathbf{v}}_j$ share the same template vertex index and UV parameterization, their correspondence is established directly, without nearest-neighbor matching or ICP. We estimate a similarity transform from the supported correspondences $\{(\hat{\mathbf{v}}_{i,j}, \bar{\mathbf{v}}_j) \mid j \in \mathcal{J}_i\}$ using Umeyama alignment [Umeyama 1991]:

$$\mathcal{T}_i(\mathbf{x}) = s_i \mathbf{R}_i^c \mathbf{x} + \mathbf{t}_i^c, \tag{11}$$

where s_i, $\mathbf{R}_i^c$, and $\mathbf{t}_i^c$ denote the estimated scale, rotation, and translation, respectively. We then apply the transform to all valid texels:

$$\mathbf{P}_i^{uv}(\mathbf{q}) = \mathcal{T}_i\left(\widetilde{\mathbf{P}}_i^{uv}(\mathbf{q})\right), \qquad \text{for } \widetilde{\mathbf{A}}_i^{uv}(\mathbf{q}) = 1. \tag{12}$$

The valid mask is kept unchanged except that it is restricted to the valid UV region of the template.

After this step, each input view is represented as a canonical partial UV-space point map:

$$\left\{\left(\mathbf{P}_i^{uv}, \mathbf{A}_i^{uv}\right)\right\}_{i=1}^{N}. \tag{13}$$

These canonicalized partial observations serve as the input to our UV-space multi-view fusion network.

3.3 Mask-aware Neural Fusion in UV Space

After image-to-UV lifting, each input view is represented as a canonical partial UV-space point map $\mathbf{P}_i^{uv}$ with a corresponding valid mask $\mathbf{A}_i^{uv}$. Although these partial observations are aligned to the same canonical UV domain, they remain incomplete and noisy due to self-occlusion, inaccurate UV correspondence, and residual errors in the predicted point maps. We therefore introduce a neural fusion network to aggregate these partial observations and predict a complete UV-space point map.

Shared per-view encoding. A straightforward solution is to directly average the partial point maps in UV space. However, raw averaging collapses noisy and incomplete observations before the network can reason about their reliability and local context. Instead, we first encode each view independently with a shared encoder. For the i-th view, we concatenate the canonical UV point map and its valid mask as input:

$$\mathbf{F}_i = E_\theta\left(\mathbf{P}_i^{uv} \oplus \mathbf{A}_i^{uv}\right), \tag{14}$$

where E_θ is a convolutional encoder shared by all views, $\oplus$ denotes channel-wise concatenation, and $\mathbf{F}_i$ is the encoded feature map of the i-th partial observation.

The shared encoder has two important roles. First, it extracts local geometric context from each partial UV point map before cross-view aggregation. Second, because the same encoder is applied to all views, the representation is independent of the input view ordering and can naturally handle a variable number of views.

Mask-aware feature fusion. Since each UV point map only covers a subset of the face, invalid texels should not contribute to the fused representation. We therefore aggregate the encoded features using the corresponding valid masks. Let $\mathbf{A}_i^f$ denote the valid mask resized to the spatial resolution of $\mathbf{F}_i$. The fused feature map is computed by masked averaging:

$$\bar{\mathbf{F}}(\mathbf{q}) = \frac{\sum_{i=1}^{N} \mathbf{A}_i^f(\mathbf{q})\, \mathbf{F}_i(\mathbf{q})}{\sum_{i=1}^{N} \mathbf{A}_i^f(\mathbf{q}) + \epsilon}, \tag{15}$$

where $\mathbf{q}$ indexes a spatial location in the feature map and ϵ is a small constant for numerical stability. This operation only aggregates features from views that provide valid evidence at the corresponding UV location. When multiple views observe the same region, their encoded features are averaged; when a region is visible in only one view, the fused feature is determined by that view; and when no view observes a region, the decoder must infer the missing geometry from surrounding context and the learned facial prior.

UV point map completion. The fused feature map is then passed to a decoder to predict a complete UV-space point map:

$$\mathbf{P}^{uv} = D_\theta\left(\bar{\mathbf{F}}\right), \tag{16}$$

where D_θ is a convolutional decoder and $\mathbf{P}^{uv} \in \mathbb{R}^{S\times S\times 3}$ is the reconstruction point map. The decoder completes missing UV regions and regularizes noisy observations into a coherent facial surface. The fixed-topology mesh is obtained by sampling $\mathbf{P}^{uv}$ at the template vertex UV coordinates.

Relation to raw UV averaging. Our design differs from a simple raw-fusion baseline that first averages the canonical UV point maps and then feeds the averaged point map to a decoder. This baseline performs fusion directly in coordinate space and therefore loses view-specific information before neural processing. In contrast, our network performs fusion in feature space: each partial observation is first interpreted by the shared encoder, and only then aggregated using valid masks. This allows the network to denoise individual observations, exploit local UV context, and learn how to complete unobserved regions more effectively.

3.4 Training Objectives

For each training subject, we bake the ground-truth fixed-topology mesh into the canonical UV domain to obtain a ground-truth UV point map $\mathbf{P}_{gt}^{uv} \in \mathbb{R}^{S\times S\times 3}$ and its valid mask $\mathbf{A}_{gt}^{uv}$. Given the predicted reconstruction point map $\mathbf{P}^{uv}$, we use a masked L2 reconstruction loss in UV space:

$$\mathcal{L}_{pm} = \frac{\sum_{\mathbf{q}} \mathbf{A}_{gt}^{uv}(\mathbf{q}) \left\| \mathbf{P}^{uv}(\mathbf{q}) - \mathbf{P}_{gt}^{uv}(\mathbf{q}) \right\|_2^2}{\sum_{\mathbf{q}} \mathbf{A}_{gt}^{uv}(\mathbf{q}) + \epsilon}. \tag{17}$$

Since the final output is a fixed-topology mesh, we also supervise the vertices sampled from the predicted UV point map. Let $\mathbf{V}_{recon}$ and $\mathbf{V}_{gt}$ denote the predicted and ground-truth mesh vertices, respectively. The vertex loss is

$$\mathcal{L}_v = \frac{1}{M} \sum_{j=1}^{M} \left\| \mathbf{v}_{recon,j} - \mathbf{v}_{gt,j} \right\|_2^2. \tag{18}$$

To reduce local surface artifacts, we further match the Laplacian coordinates of the predicted and ground-truth meshes. Let $\mathbf{L}$ be the uniform mesh Laplacian constructed from the fixed topology $\mathbf{F}$. The Laplacian matching loss is

$$\mathcal{L}_{lap} = \frac{1}{M}\sum_{j=1}^{M}\left\|(\mathbf{L}\mathbf{V}_{recon})_j - (\mathbf{L}\mathbf{V}_{gt})_j\right\|_1. \tag{19}$$

This term encourages the reconstructed mesh to preserve the local differential structure of the ground-truth surface, which helps suppress spurious bumpy artifacts.

The full objective for the neural fusion network is

$$\mathcal{L} = \lambda_{pm}\mathcal{L}_{pm} + \lambda_v\mathcal{L}_v + \lambda_{lap}\mathcal{L}_{lap}. \tag{20}$$

In all experiments, we set $\lambda_{\text{pm}} = 0.1$, $\lambda_{\text{v}} = 10$, and $\lambda_{\text{lap}} = 1$, and use the same weights throughout training.

4 EXPERIMENTS

4.1 Experimental Setup

Implementation details. All input images are resized to 512×512, and both the partial and complete UV-space point maps are represented at a resolution of 512×512. We use the official pretrained VGGT and Pixel3DMM models and keep their parameters frozen. We train on the official ground-truth meshes of Ava-256. During each training iteration, we randomly sample between 4 and 16 input views for each example. The network is optimized using AdamW with a learning rate of 5×10^{-3}. The global batch size is 4. We train for 200 epochs on two RTX 4090 GPUs, which takes approximately two days. Since the official Ava-256 meshes share a common topology, we manually specify sparse correspondences between the FLAME and Ava-256 templates only once and construct a reusable automatic registration pipeline in Wrap[1]. More details are provided in the supplementary material.

Datasets. We evaluate our method on four datasets: H3DS [Caselles et al. 2025; Ramon et al. 2021], NeRSemble [Kirschstein et al. 2023], EmoTalk3D [He et al. 2024], and RenderMe-360 [Pan et al. 2024]. We additionally capture several in-the-wild multi-view sequences for evaluation. Unless otherwise stated, all methods are evaluated with 16 input views. For EmoTalk3D, we use 11 views since only 11 camera views are available. We also report 4-view quantitative and qualitative results in the supplementary material.

Baselines. We compare with both monocular and multi-view face reconstruction methods. The monocular baselines include DECA [Feng et al. 2021], 3DDFA-V3 [Wang et al. 2024b], and Pixel3DMM [Giebenhain et al. 2026]. For these methods, we run inference on all input views and report the best reconstruction. The multi-view baselines include DFNRMVS [Bai et al. 2020], HRN [Lei et al. 2023], and VGGTFace [Ming et al. 2026]. We also include an ablation variant of our method that removes neural feature fusion: it directly computes a masked average of the canonical UV-space point maps and feeds the averaged point map to the same decoder.

Metrics. We use Chamfer distance as the main quantitative metric and report all errors in millimeters. For H3DS and NeRSemble, we follow the evaluation protocol used by prior work and quote the numbers reported in SIRA++ [Caselles et al. 2025] and VGGTFace. For EmoTalk3D and RenderMe-360, we reconstruct ground-truth geometry with Metashape[2] using all available camera views provided by the datasets. Following the evaluation protocol of SIRA++, we align each predicted mesh to the corresponding ground-truth mesh using a similarity transformation followed by rigid ICP before computing the Chamfer distance. We also compare runtime across methods. For our method, the reported runtime includes VGGT inference, Pixel3DMM inference, UV-space point-map construction, neural fusion network inference, and final mesh extraction. All runtime measurements are conducted on a single RTX 4090 GPU.

4.2 Quantitative Comparison

Table 1 reports the quantitative comparison on four evaluation datasets. Our method achieves the lowest Chamfer distance on almost all reported metrics, including the mean error on H3DS and the mean/median errors on NeRSemble, EmoTalk3D, and RenderMe-360. Compared with monocular methods, our method benefits from multi-view geometric observations and produces more accurate fixed-topology reconstructions. Compared with multi-view baselines, our canonical UV-space representation more effectively aggregates partial observations from different views.

Table 1 also reports the runtime of different methods. Our full pipeline takes less than 3 seconds on a single RTX 4090. While some monocular methods are faster, they use only a single image and have noticeably higher reconstruction errors. Among multi-view methods, our method achieves both higher accuracy and faster inference, without requiring per-subject optimization.

4.3 Qualitative Comparison

Figure 7 and Figure 8 show qualitative comparisons on public datasets and in-the-wild captures. Monocular methods, such as Pixel3DMM and 3DDFA-V3, can recover plausible coarse face shapes, but their reconstructions are often biased toward generic facial priors. As a result, they tend to miss subject-specific traits and struggle with challenging expressions, such as tightly closed eyes, puffed cheeks, asymmetric smiles, and large mouth deformations. Multi-view methods improve geometric consistency by using more observations, but DFNRMVS and HRN can still produce noisy surfaces, distorted local structures, or inaccurate expression geometry.

Compared with these baselines, our method reconstructs cleaner fixed-topology meshes with more faithful global shape and local facial deformation. In particular, our results better preserve expression-related geometry around the eyes, mouth, cheeks, and nasolabial regions, while avoiding the surface noise and local artifacts produced by some multi-view baselines. VGGTFace benefits from strong VGGT geometry and produces competitive results, but its heuristic fusion can over-smooth local facial structures or fail to fully resolve inconsistent observations. The in-the-wild results further demonstrate the generalization ability of our approach. Although

[1] https://faceform.com

[2] https://www.agisoftmetashape.com

Table 1. Quantitative comparison on public datasets (all metrics in mm, ↓ indicates lower is better).

Method	H3DS	NeRSemble			EmoTalk3D			RenderMe-360			Time	# Views
	Mean ↓	Mean ↓	Median ↓	Std ↓	Mean ↓	Median ↓	Std ↓	Mean ↓	Median ↓	Std ↓		
DECA	1.99	1.58	1.55	1.18	1.56	1.51	1.23	1.71	1.65	1.24	< 1*s*	1
3DDFA-V3	1.81	1.52	1.46	1.08	1.51	1.46	1.19	1.72	1.65	1.27	< 1*s*	1
Pixel3DMM	1.69	1.42	1.38	1.01	1.31	1.28	0.99	1.48	1.38	1.06	~ 60*s*	1
DFNRMVS	1.78	1.62	1.58	1.09	1.72	1.67	1.30	1.77	1.68	1.25	~ 30*s*	16
HRN	1.46	1.49	1.43	1.10	1.64	1.57	1.28	1.65	1.55	1.30	~ 70*s*	16
VGGTFace	1.18	0.98	0.98	**0.65**	1.02	0.99	0.76	1.08	0.98	0.74	< 10*s*	16
Ours	**1.12**	**0.91**	**0.91**	0.69	**0.89**	**0.86**	**0.74**	**0.93**	**0.84**	**0.72**	< 3*s*	16
w/o Neural Fusion	1.22	1.01	0.99	0.76	1.01	0.98	0.82	1.03	0.91	0.83	< 3*s*	16

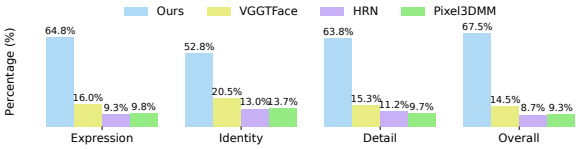

Fig. 3. User study results. Participants compared the reconstruction results of Pixel3DMM, HRN, VGGTFace, and our method without seeing the method names. Our method receives the highest preference across expression matching, identity preservation, detail preservation, and overall quality.

trained on Ava-256, our method remains robust under casual capture conditions with different lighting, viewpoints, identities, and expressions.

4.4 User Study

To further evaluate the perceptual quality of the reconstructed meshes, we conducted a user study comparing our method with Pixel3DMM, HRN, and VGGTFace. Each participant was shown the input multi-view images and the reconstructed meshes produced by the four methods. The method names were hidden during the study. For each example, the four results were randomly assigned to options A, B, C, and D to avoid ordering bias.

For each test case, participants were asked to answer the following four questions: (1) Which reconstruction best matches the facial expression in the input images? (2) Which reconstruction best matches the identity-related facial features, such as face shape and facial components? (3) Which reconstruction best preserves the facial details in the input images? (4) Overall, which reconstruction result is the best? We provide a screenshot of the web interface used in our user study in the supplementary material.

We invited 25 participants, and each participant evaluated 24 groups of reconstruction results. Therefore, each criterion received 600 votes in total. The final voting percentages are shown in Figure 3. Our method is preferred by the majority of participants across all four criteria. These results indicate that the advantages of our method are not only reflected in numerical errors, but are also clearly perceived by human observers.

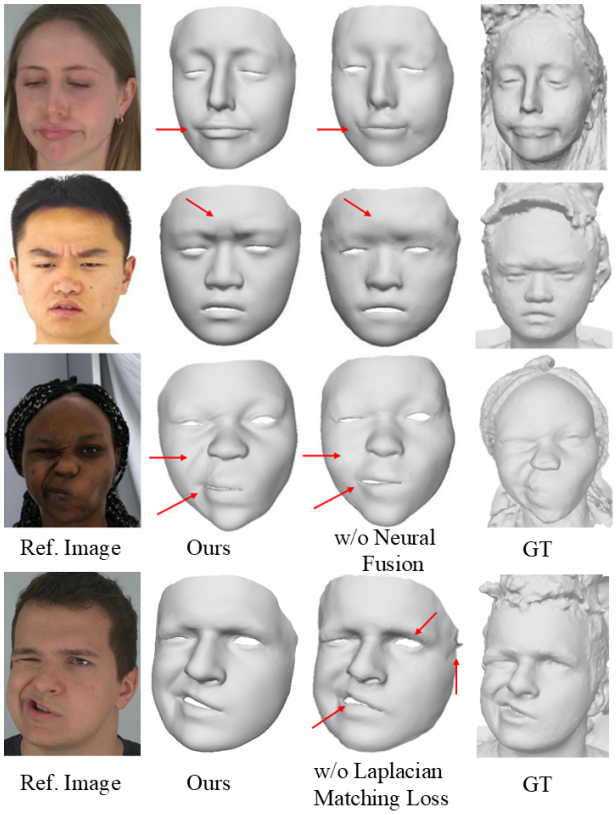

Fig. 4. Qualitative evaluation of the key design choices in our method. Without neural fusion, directly averaging canonical UV-space point maps loses view-specific evidence before decoding, leading to less accurate expressions and degraded local facial geometry. Without the Laplacian matching loss, the reconstructed mesh captures the overall shape but exhibits less regular surfaces and visible bumpy artifacts.

4.5 Ablation Studies

We analyze several important design choices in our method: neural feature fusion, mask-aware fusion and Laplacian matching. As shown in Table 1, removing neural fusion and directly averaging

Table 2. Ablation of the input and fusion masks. Relative errors are averaged over the four public benchmarks and normalized to the full model.

Variant	Input Mask	Fusion Mask	Relative Error ↓
Full model	✓	✓	1.00
w/o input mask		✓	1.00
w/o fusion mask	✓		1.06
w/o both masks			1.06

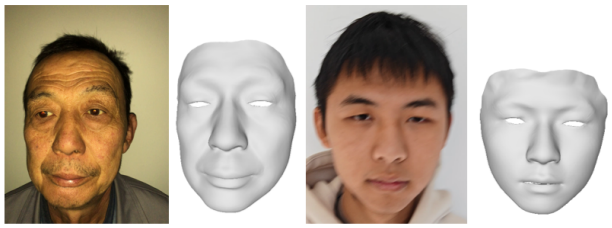

Fig. 5. Qualitative robustness on challenging real captures. Each pair shows one representative input image and the reconstruction obtained from the complete multi-view sequence. Our method remains stable under challenging illumination and motion blur.

canonical UV-space point maps leads to worse quantitative performance on all four datasets. This suggests that raw coordinate-space averaging is insufficient for handling noisy and partially inconsistent observations.

Figure 4 further shows the qualitative effect of this design. Without neural fusion, the decoder receives an already-collapsed UV point map, where view-specific evidence and local reliability cues have been lost. As a result, the reconstructed mesh can have less accurate expressions and weaker local geometry.

We further ablate the two uses of the validity mask: the *input mask*, concatenated with each partial XYZ map before encoding, and the *fusion mask*, used to suppress invalid observations during cross-view feature aggregation. Table 2 reports the relative Chamfer error averaged over the four public benchmarks and normalized to the full model. Removing the fusion mask increases the error by 6%, with or without the input mask, whereas removing only the input mask causes no measurable degradation. This confirms that validity-aware feature aggregation is the key component, while zero-valued invalid texels already provide a consistent cue to the encoder. We nevertheless retain both masks in the full model, as the input mask explicitly distinguishes missing observations from valid geometry whose normalized coordinates may be close to zero.

We also ablate the Laplacian matching loss. Without this term, the reconstructed mesh still captures the overall face shape, but the surface becomes less regular and exhibits visible bumpy artifacts.

4.6 Robustness Evaluation

To evaluate robustness beyond standard capture conditions, we consider two complementary settings: challenging real captures and controlled stress tests on public benchmarks. Figure 5 presents representative real-capture results under difficult illumination and motion blur caused by rapid hand-held camera movement. Our method also tolerates transient inconsistencies during a short sequential capture, such as blinking or occasional brief expression changes. Such variations typically affect only a small subset of views, while the remaining observations still support a dominant facial state. Since our fusion network operates on the geometric predictions rather than directly on RGB appearance, image degradations influence the final reconstruction primarily through their effects on the upstream predictors. Persistent or large expression changes, however, violate our static-capture assumption.

Table 3. Robustness stress tests. Within each row, the settings and relative errors are listed in the same order. Detailed corruption protocols are provided in the supplementary material.

Stress setting	Relative error ↓
Low / over / side / point lighting	1.11/1.03/1.01/1.00
Motion blur on 2/4/8 frames	1.01/1.02/1.05
Temporal replacement of 1/2/4 frames	1.00/1.01/1.02
8-view: side A / side B / profile-heavy	1.08/1.08/1.06
4-view: side A / side B / profile-heavy	1.39/1.38/1.25
4-view retrained: side A / side B / profile-heavy	1.26/1.25/1.22
VGGT: Gaussian / smooth / local outliers	1.02/1.15/1.13
Pixel3DMM: jitter / smooth warp / local corruption	1.03/1.12/1.13

We further quantify these observations through controlled stress tests covering illumination changes, motion blur, mild temporal inconsistency, imbalanced view selection, and errors from VGGT and Pixel3DMM. For image-space corruptions, we modify the RGB inputs and rerun the complete pipeline. For upstream corruptions, we perturb the predicted point maps or UV maps before UV-space lifting. Table 3 reports the relative Chamfer error averaged over the four public benchmarks and normalized by the corresponding clean full-view result.

Overall, moderate appearance and temporal disturbances cause only limited degradation, indicating that our method remains stable under motion blur and mild temporal inconsistency during sequential capture. Most lighting perturbations also remain close to the clean setting, although severe low light is more challenging. Eight-view one-sided and profile-heavy inputs remain stable, whereas four-view inputs that completely omit one facial side are substantially harder; view-count-specific retraining partially reduces this degradation. The method is also tolerant to moderate random upstream noise, while spatially coherent point-map deformation and systematic UV-correspondence errors have a larger impact. These results are consistent with the qualitative real-capture observations and indicate graceful degradation under moderate or isolated corruptions. Detailed corruption protocols and additional visualizations are provided in the supplementary material.

4.7 Limitations and Discussions

Figure 6 shows typical failure cases of our method. First, our reconstruction may include bangs or hair-like geometry near the forehead. This happens because both the VGGT point maps and the registered training meshes may preserve such geometry, causing the fixed-topology face surface to absorb the bangs into the reconstructed

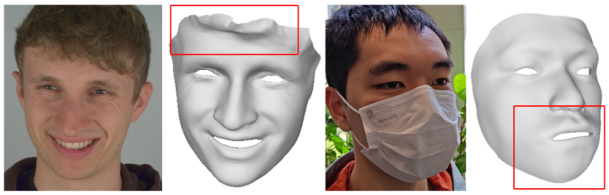

Fig. 6. Limitations of our method. Our method can absorb bangs into the facial mesh when hair geometry is present in the predicted point maps or registered training meshes. It also struggles with severe occlusions such as masks, where UV correspondences become unreliable and the occluder is reconstructed as part of the visible geometry.

mesh. Second, our method struggles with strong occlusions such as face masks. In these cases, Pixel3DMM may provide inaccurate UV correspondences around the occluded region, while VGGT reconstructs the occluder as part of the visible geometry. Since our representation does not explicitly separate occluders from the face surface, the final reconstruction can fail under severe occlusion. Incorporating explicit face–hair and face–occluder decomposition could prevent non-face geometry from being absorbed into the fixed-topology facial surface. In addition, our current system reconstructs each capture as a static subject. Extending the proposed canonical UV-space fusion framework to dynamic sequences, with temporal consistency and motion-aware fusion, would be an interesting direction. We provide further discussions of limitations in the supplementary material.

5 CONCLUSIONS

We presented UVFaceFusion, a feed-forward framework for fixed-topology face reconstruction from in-the-wild multi-view images. By lifting VGGT point maps into a canonical facial UV space with Pixel3DMM correspondences, our method turns topologically consistent reconstruction into UV point-map completion and replaces heuristic optimization with mask-aware neural fusion. Experiments demonstrate our method obtains state-of-the-art results on various benchmarks and strong generalization ability on in-the-wild data captured by everyday users.

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Fig. 7. **Qualitative comparisons on public datasets.** Our method reconstructs cleaner fixed-topology meshes and better preserves challenging expression geometry around the eyes, mouth, and cheeks. Error maps are shown beside each reconstruction.

Fig. 8. **Qualitative comparisons on in-the-wild captures.** Our method generalizes to casual multi-view images and produces stable, faithful face meshes across diverse identities, viewpoints, lighting conditions, and expressions. Error maps are shown beside each reconstruction.

UVFaceFusion: Fast Multi-view Topologically Consistent Face Reconstruction in the Wild via UV-space Neural Fusion *Supplementary Material*

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A Quantitative Evaluation with Four Input Views

In the main paper, we report quantitative comparisons under the 16-view input setting. To further evaluate the performance of our method under a more sparse-view scenario, we conduct an additional quantitative experiment using only four input views. The evaluation protocol, metrics, and test datasets are the same as those used in the main paper. The results are reported in Table 1. Lower values indicate better reconstruction accuracy.

As shown in Table 1, our method consistently achieves the best performance across all evaluated datasets under the 4-view setting. The results demonstrate that our method remains effective even when the number of input views is significantly reduced. This robustness mainly benefits from our UV-space fusion strategy, which aggregates partial observations from different views using valid masks, and from the learned shape prior that completes unobserved regions in the canonical UV domain. Moreover, our method runs at approximately 1 second per subject, which is faster than the compared methods while producing more accurate reconstructions.

B Qualitative Evaluation with Four Input Views

We further provide qualitative comparisons under the 4-view input setting in Fig. 7. The examples include both public benchmark datasets, namely H3DS, NeRSemble, EmoTalk3D, and RenderMe-360, and in-the-wild captures collected by ourselves. All methods are evaluated using the same sparse-view inputs.

The 4-view setting is more challenging than the 16-view setting because large portions of the face may be weakly observed or unobserved. As a result, directly fusing sparse observations or relying on per-view predictions can easily lead to incomplete geometry, inaccurate facial expressions, or over-smoothed surfaces. In contrast, our method reconstructs a complete fixed-topology face mesh by completing the partial UV-space point maps in a canonical domain. The qualitative results show that our method better preserves the input expression and identity-related facial geometry, while producing more stable and detailed surfaces. These results are consistent with the quantitative improvements reported in Table 1.

C More Details on Our Captured Data

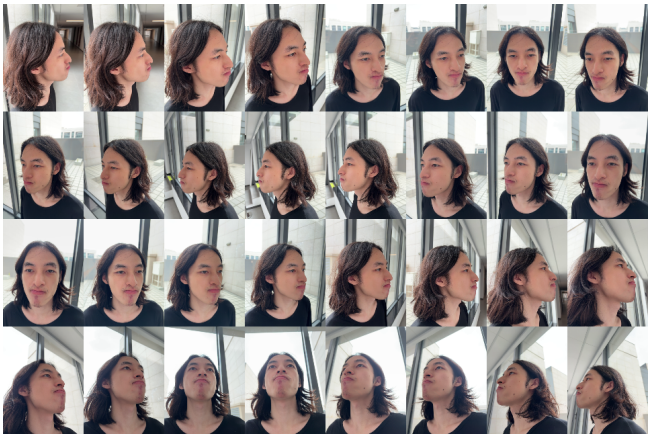

Fig. 1. Montage of densely sampled frames from one of our self-captured in-the-wild videos.

We provide additional details about our self-captured in-the-wild data. The videos are recorded with a hand-held smartphone, including both iPhone and Android devices, at a resolution of 1920×1080. For each subject, we capture a short video by moving the phone around the subject's head at a distance of approximately 0.8–1.5 meters. For the 16-view setting, we uniformly sample 16 frames from each video and use them as the multi-view input to our system. For visualization, Fig. 1 shows a denser sampling of frames from one captured video, illustrating the camera trajectory and the viewpoint coverage during capture.

D Monocular Reconstruction

Although our method is designed for multi-view face reconstruction, it can also be naturally extended to the monocular setting. VGGT can predict a reasonable point map from a single input image. We then

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Table 1. Quantitative comparison using four input views. Lower values are better. Our method consistently achieves the lowest reconstruction errors across all datasets while maintaining the fastest inference speed.

Method	H3DS	NeRSemble			EmoTalk3D			RenderMe-360			Time	# Views
	Mean ↓	Mean ↓	Median ↓	Std ↓	Mean ↓	Median ↓	Std ↓	Mean ↓	Median ↓	Std ↓		
DFNRMVS	1.78	1.63	1.58	1.10	1.74	1.67	1.31	1.80	1.71	1.26	~ 5*s*	4
HRN	1.56	1.57	1.56	1.16	1.73	1.61	1.34	1.83	1.74	1.36	~ 15*s*	4
VGGTFace	1.37	1.09	1.05	0.80	1.21	1.11	0.90	1.28	1.15	0.91	~ 3*s*	4
Ours	**1.31**	**1.06**	**1.02**	**0.74**	**1.02**	**1.01**	**0.73**	**1.12**	**1.04**	**0.84**	~ 1*s*	4

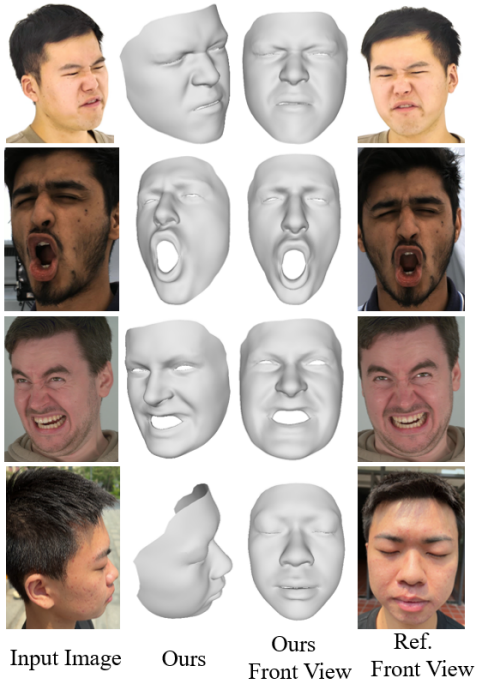

Fig. 2. Monocular reconstruction results. Although our method is mainly designed for multi-view inputs, it can be extended to the single-view setting by using one partial UV-space point map and completing the missing regions in the canonical UV domain. These results demonstrate the flexibility of our formulation. Note, our method only uses the input image in the first column; the front-view image in the last column is shown only as a reference and is not used for reconstruction.

lift the predicted point map into the UV space using the UV map estimated by Pixel3DMM, obtaining a partial UV-space point map and its corresponding valid mask. In this case, our reconstruction network takes only one partial UV-space point map as input, and completes the missing regions using the learned facial shape prior in the canonical UV domain.

We show several monocular reconstruction results in Fig. 2. In the monocular setting, our method takes only the input image as input, while the front-view image is provided solely for visual reference and is not used by our method. Since monocular reconstruction is not the main focus of this work, we do not include quantitative comparisons or baseline comparisons in this setting. Nevertheless, the results demonstrate that our UV-space formulation is flexible and can handle extremely sparse observations. Even with only a single view, our method can reconstruct a complete fixed-topology face mesh and produce plausible geometry for the unobserved regions.

E View-Number Sensitivity

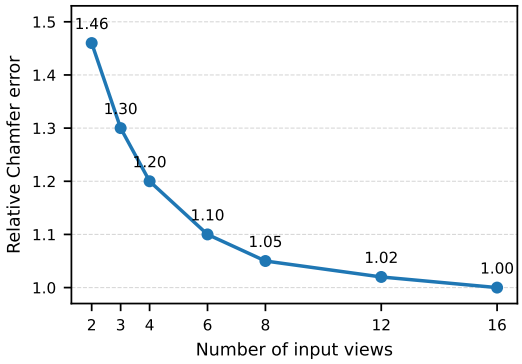

Fig. 3. View-number sensitivity of our method. We report the mean relative Chamfer error over the four public benchmarks, normalized by the corresponding full-view result. The reconstruction error increases gradually as the number of input views decreases.

We evaluate the same trained model with 16,12,8,6,4,3 and 2 input views. For each setting, only the number of input views is changed, without view-count-specific fine-tuning. We report the mean relative Chamfer error over the same four public benchmarks used in the main paper, with the corresponding full-view result normalized to 1.00. As shown in Fig. 3, the relative errors for 16,12,8,6,4,3 and 2 views are 1.00,1.02,1.05,1.10,1.20,1.30 and 1.46, respectively. Performance degrades smoothly as observations become sparser. Although the network is trained with randomly sampled 4-16 views, it also produces reasonable reconstructions with two or three views, which are outside its training view-count range. These results demonstrate that the proposed fusion framework is not tied to exactly 16 inputs, while also confirming that extremely sparse observations remain substantially more challenging.

F Robustness Stress Tests

We conduct controlled stress tests on the public benchmarks to quantify robustness under common capture and upstream-prediction failures. For image corruptions, we modify the original RGB inputs and rerun the entire reconstruction pipeline. We report relative Chamfer errors averaged over the four public benchmarks and normalized by the corresponding clean, full-view result. Representative corrupted inputs are shown in Fig. 4.

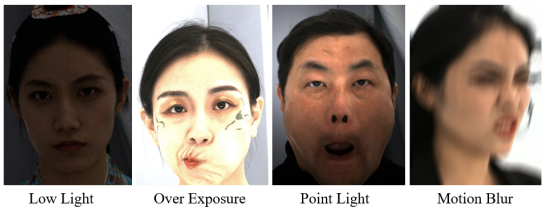

Fig. 4. Representative synthetic input corruptions used in our robustness evaluation. The corruptions are applied to the RGB images before rerunning the complete reconstruction pipeline.

Lighting. We test low light, overexposure, side lighting, and localized point lighting. Under severe low light, simulated using an exposure reduction of 3.5 EV together with image noise, the relative error is 1.11. Overexposure, side lighting, and point lighting yield relative errors of 1.03, 1.01, and 1.00, respectively. These results indicate that the full pipeline remains stable under substantial illumination variation.

Motion blur. We apply severe motion blur with a kernel length of 35 pixels to 2, 4, or 8 of the 16 input frames. The corresponding relative errors are 1.01, 1.02, and 1.05. Thus, occasional blurred frames have little effect, while the error increases moderately when a large fraction of the observations is corrupted.

Sequential-capture inconsistency. To simulate mild temporal variation during sequential capture, we randomly replace 1, 2, or 4 input frames with frames captured approximately one second earlier or later. The resulting relative errors are 1.00, 1.01, and 1.02, respectively. This result is consistent with our real captures, where blinking and occasional brief expression changes cause little visible degradation. However, persistent expression changes across many views violate the dominant-static-geometry assumption of the current method.

View imbalance. We additionally evaluate one-sided and profile-dominated view selections. For 8-view inputs, selecting views from only one facial side, from the opposite side, or using a profile-heavy set produces relative errors of 1.08, 1.08, and 1.06, respectively, compared with the full 16-view setting. With only 4 views, the errors increase to 1.39, 1.38, and 1.25. Retraining the network specifically for 4-view inputs reduces them to 1.26, 1.25, and 1.22, showing that moderate imbalance is handled well, whereas sparse inputs that completely omit one facial side remain substantially more difficult.

Upstream errors. We separately perturb the VGGT point maps and Pixel3DMM UV maps before UV-space lifting. For VGGT, Gaussian noise, smooth geometry deformation, and local outlier patches produce relative errors of 1.02, 1.15, and 1.13, respectively. For Pixel3DMM, random UV jitter, smooth UV warping, and local correspondence corruption yield 1.03, 1.12, and 1.13. The network therefore tolerates moderate unstructured noise, while spatially coherent geometry errors and systematically corrupted UV correspondences are more harmful because they introduce plausible but incorrect facial evidence.

G Further Discussion of Limitations

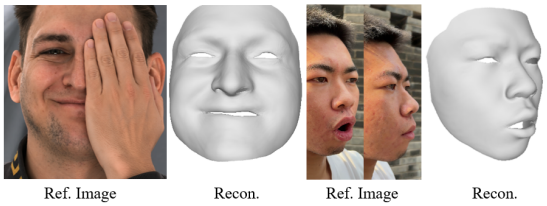

Fig. 5. Failure cases under persistent occlusion and expression changes. Left: when one side of the face is occluded by a hand, the visible region remains plausible, while the persistently hidden region loses subject-specific geometry. Right: when the subject repeatedly opens and closes the mouth during capture, different views observe incompatible facial states, causing the reconstructed mouth to converge to an intermediate configuration between the open and closed states.

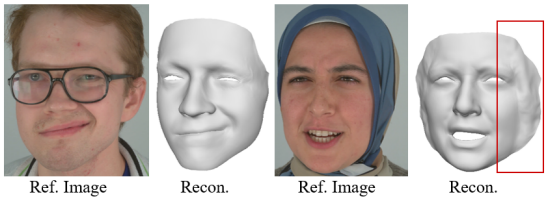

Fig. 6. Additional limitations of our method. Left: since eyeglasses are not represented in the registered training meshes, our method predicts the underlying bare-face geometry rather than reconstructing the glasses. Right: non-face geometry close to the face, such as hair or headwear, may be absorbed into the fixed-topology facial surface, as highlighted by the red box.

Severe occlusion. Our method cannot reliably recover subject-specific geometry that remains occluded throughout the entire multi-view capture. To quantify this limitation, we add either a face mask or a left-hand occluder covering one side of the face to every input image. The face-mask experiment produces relative errors of 1.92 and 1.05 in the occluded and non-occluded regions, respectively, while the hand-occlusion experiment produces 1.98 and 1.06. As illustrated in the left part of Fig. 5, the visible facial region remains plausible, whereas the persistently hidden region loses subject-specific geometry and becomes distorted. This occurs

because VGGT may interpret parts of the occluder as visible geometry, while Pixel3DMM may provide unreliable UV correspondences around the occlusion boundary.

Expression changes. The current method assumes that all input views observe a single dominant facial geometry. Blinking and occasional brief expression changes generally affect only a small number of frames and therefore have little influence when most observations remain mutually consistent. However, persistent expression changes violate this assumption. As shown in the right part of Fig. 5, we ask the subject to repeatedly open and close the mouth during capture. Since different views observe substantially different mouth states, the reconstructed mouth lies between the open and closed configurations and does not accurately match either state. This demonstrates that fusing incompatible non-rigid geometry across views can produce averaged or locally inconsistent facial structures. Handling such captures would require temporal correspondence, motion-aware fusion, or grouping the input frames according to facial state.

Hair and eyeglasses. As illustrated in Fig. 6, eyeglasses are absent from the registered training meshes, and the network therefore tends to reconstruct the underlying bare face rather than the glasses geometry. Conversely, hair, bangs, or closely attached headwear may be absorbed into the reconstructed surface when they are preserved in the upstream point maps or registered training geometry. Explicit face-hair, face-accessory, and face-occluder decomposition would help prevent non-face geometry from being incorporated into the facial mesh.

Upstream failures. Our fusion network can regularize moderate random errors, but it remains dependent on the geometry and correspondences provided by VGGT and Pixel3DMM. Spatially coherent point-map deformation or systematically incorrect UV correspondences may appear as plausible observations and survive multi-view fusion. In particular, the validity masks indicate whether a UV location is observed, but do not guarantee that the associated geometry or correspondence is correct. Such errors may affect both per-view canonicalization and the final reconstructed surface. Incorporating calibrated upstream confidence, robust outlier rejection, or joint refinement of geometry and correspondences is an important direction for future work.

H Additional Evaluation on NoW

Our method reconstructs the observed facial geometry without explicitly disentangling identity and expression. Since NoW evaluates reconstructions against neutral ground-truth scans, expressive inputs would conflate expression differences with reconstruction errors. We therefore evaluate our method only on neutral-expression scenes. Using all 2–3 available images per scene, our main multi-view model achieves a mean error of 0.98 mm, a median error of 0.78 mm, and a standard deviation of 0.67 mm. The supplementary single-view variant obtains 1.19 mm, 0.92 mm, and 0.78 mm, respectively. These results demonstrate that our method maintains accurate reconstruction under sparse multi-view inputs, while additional views provide complementary geometric evidence.

I Multi-View Pixel3DMM Optimization Baseline

To examine whether directly combining Pixel3DMM with VGGT geometry can achieve comparable performance, we implement a multi-view Pixel3DMM optimization baseline augmented with a dense VGGT depth-consistency term. The baseline obtains mean Chamfer errors of 1.48, 1.35, 1.27, and 1.40 mm on H3DS, NeRSemble, EmoTalk3D, and RenderMe-360, respectively. These errors are 1.32×, 1.48×, 1.43×, and 1.50× those of our method. This comparison shows that adding VGGT geometric supervision to parametric multi-view optimization does not provide the same reconstruction accuracy as our learned canonical UV-space fusion.

J More Details on Canonicalization

At each template vertex UV coordinate, we perform masked local sampling within a radius-one (3×3) neighborhood using Gaussian weights with $\sigma = 0.8$. Only template vertices with sufficient valid local support (larger than 2) are retained for estimating the similarity transform. The estimated transform is applied to all valid UV texels. Invalid texels are subsequently reset to zero, and the normalized coordinates are clipped to $[-1, 1]$.

Since canonicalization applies only a global similarity transform, it does not non-rigidly deform the prediction toward the template, thereby retaining identity- and expression-dependent geometric variations. Moreover, the transform is estimated independently for each view, so reducing the total number of input views does not directly reduce the correspondence support available within an individual view. The estimation remains well constrained when sufficiently many spatially distributed facial vertices are visible. In contrast, severe occlusion or systematic errors in the VGGT point maps or Pixel3DMM UV correspondences may leave a small, spatially concentrated, or corrupted correspondence set, making the estimated transform unreliable.

K Additional Implementation Details

Network architecture. The fusion network is a five-scale U-Net with 52.99M trainable parameters. Each view is encoded independently from a four-channel input comprising its canonical XYZ map and validity mask, using shared channel widths of 32, 64, 128, 256, and 512. Each encoder and decoder stage contains two residual blocks, where each block uses two 3×3 convolutions, eight-group Group Normalization, and SiLU activations. Downsampling is performed by stride-two convolutions, while the decoder uses bilinear upsampling and skip connections. At every scale, the masks are resized by area interpolation. A 1×1 scoring network predicts per-view logits, to which we add the logarithm of the soft validity mask before applying a softmax across views. The fusion module combines the resulting observed feature, an unmasked view-mean prior, weighted feature uncertainty, and coverage using a learned gate followed by residual refinement. Two bottleneck blocks employ dilated branches with rates 1, 2, 4, 8 and a global-pooling branch. The final prediction is formulated as a residual added to the masked-average input geometry.

UV convention and masks. We use the FLAME connectivity and fixed vertex UV coordinates normalized to $[0, 1]^2$, with a coordinate (u, v) mapped to the texel location $((S-1)u, (S-1)v)$. Ground-truth

point maps and the template-valid mask are constructed using the same UV layout. Bilinear forward splatting produces the accumulated weight W_i^{uv}, and texels satisfying $W_i^{uv} > \tau_w$, where $\tau_w = 1$, are retained and intersected with the template-valid UV mask. Invalid XYZ texels are set to zero. Canonicalized input maps and ground-truth maps share the same normalized template coordinate system, with coordinates clipped to $[-1, 1]$.

Registration and optimization. Ava-256 meshes share a common native topology. We manually annotate 51 template correspondences and use them to perform automatic non-rigid registration of the FLAME template to the Ava-256 meshes in Wrap.

L User Study Screenshot

We provide a screenshot of the web interface used in our user study, as shown in Fig. 8. The left panel shows the input multi-view images and the four evaluation questions, while the right panel shows the reconstruction results from four anonymized methods, denoted as Options A–D. The four mesh viewers are interactive: participants can zoom in and drag the meshes to inspect the reconstructed geometry from different viewpoints before making their choices. The method names are not displayed, and the assignment of methods to Options A–D is randomized for each test case.

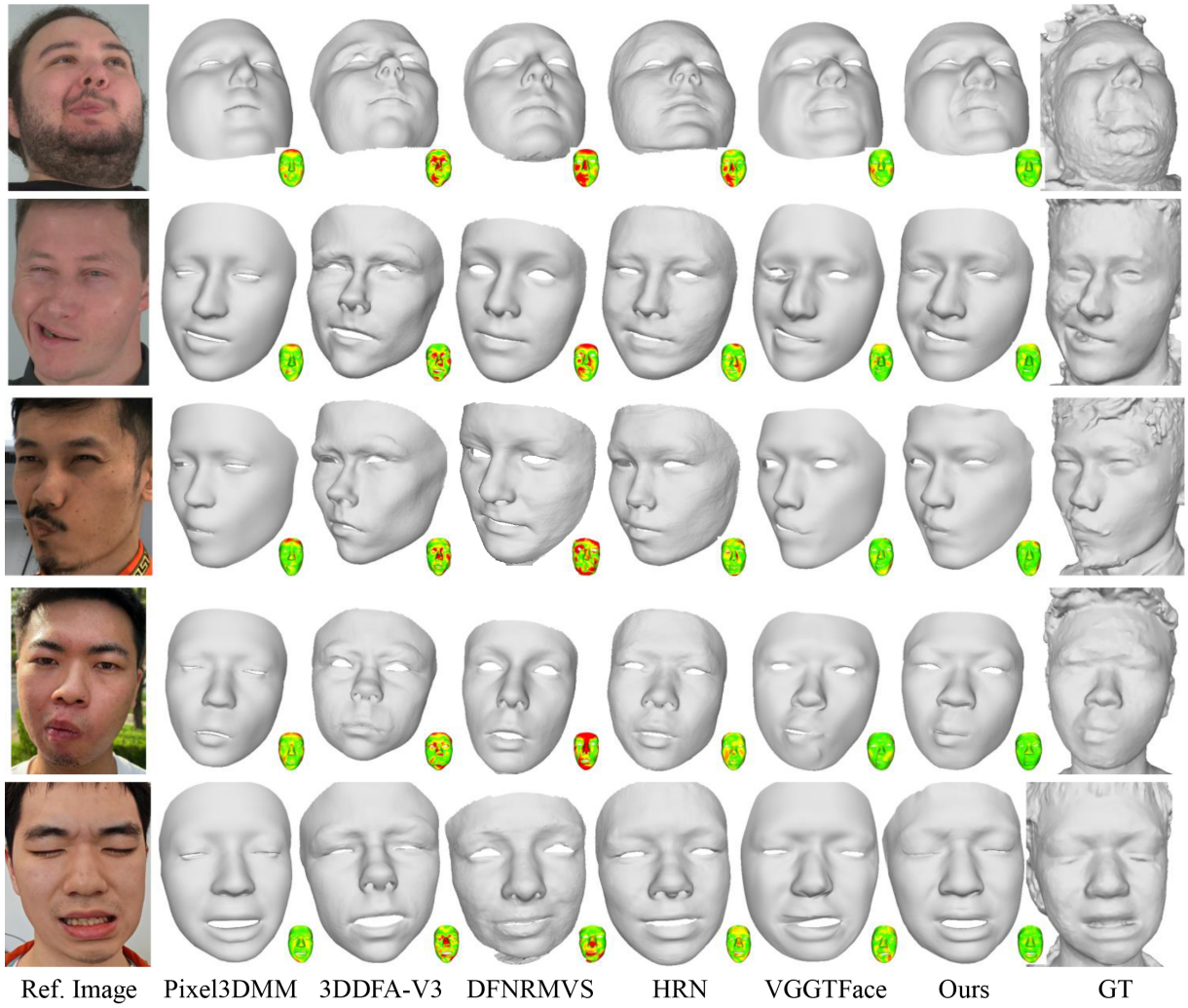

Fig. 7. Qualitative comparison under the 4-view input setting. For compactness, we show one reference image from the input views in the first column, while all methods use the same four input views. From left to right, we show the results of Pixel3DMM, 3DDFA-V3, DFNRMVS, HRN, VGGTFace, our method, and the ground truth. Compared with the baselines, our method reconstructs more accurate facial expressions and identity-related geometry, while producing more complete and stable fixed-topology meshes under sparse-view inputs.

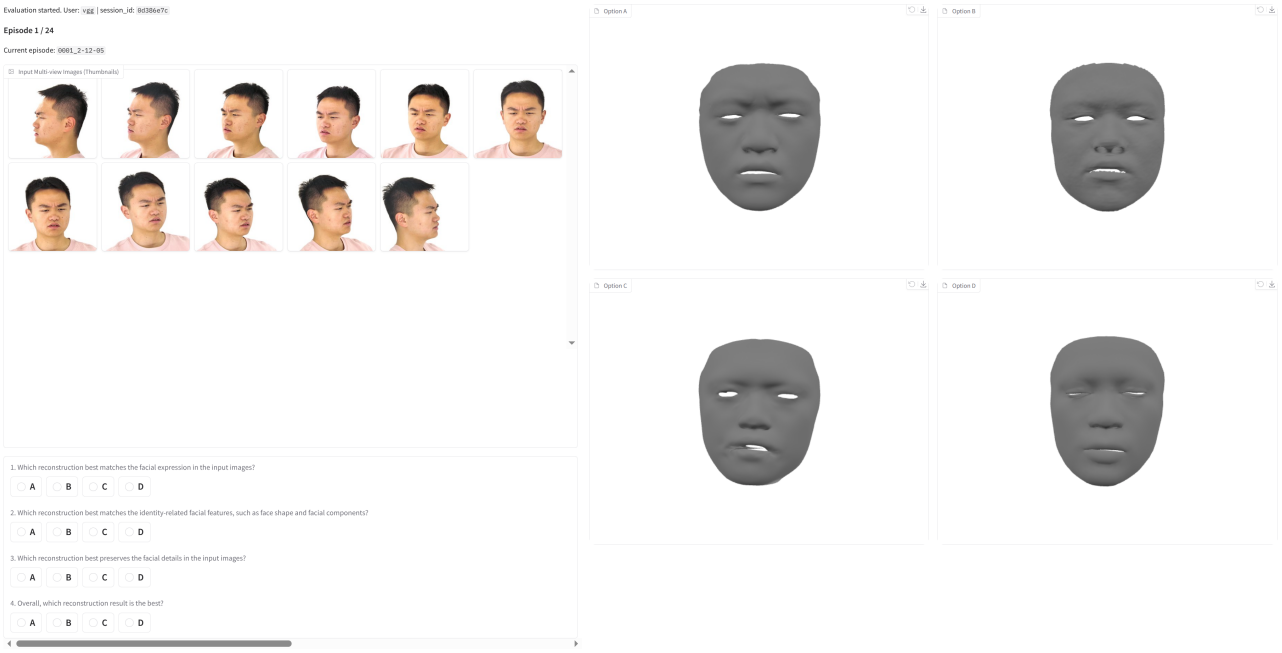

Fig. 8. Screenshot of the web interface used in our user study. Participants are shown the input multi-view images and four anonymized reconstruction results. Each mesh viewer is interactive, allowing participants to zoom in and rotate the mesh for closer inspection. Participants then select the best result for expression, identity, detail, and overall quality.